

Raymond Ferguson

CHECK POINT TASK - PSEUDOCODE

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Analysis

- Get the user's first name, surname and title and greet them based on this
- Ask if they want plastic cutlery
- Make sure all user inputs are validated (names, menu choices, allergy selections, order counts, and loyalty card options). Invalid inputs trigger clear error messages, and the user is asked to re-enter their choice.
- Display the restaurant menu
- Ask them if they have any allergies the restaurant should be aware of and don't let them select any of the menu items containing the corresponding ingredient
- Allow the user to request custom dishes at extra cost (useful for someone with allergies)
- Ask the user how many people they are ordering for and allow them to make multiple orders up to a maximum of 5, they will choose all starters then move to mains and so on
- Allow the user to skip courses if they do not want a full 3 course meal, don't want a drink or maybe just want a dessert
- After each course, display the order and ask the user if they are happy with their choice and continue on or if not, allow them to choose another option
- Ask the user for their starter choice, save this and add to data structure and if the user wishes to make multiple orders, loop back and ask again for however many orders they would like to make
- Ask the user for their main course choice, save this and add to data structure and if the user wishes to make multiple orders, loop back and ask again for however many orders they would like to make
- Ask the user for their dessert choice, save this and add to data structure and if the user wishes to make multiple orders, loop back and ask again for however many orders they would like to make
- Ask the user for their drink choice, save this and add to data structure and if the user wishes to make multiple orders, loop back and ask again for however many orders they would like to make
- Display the user's entire full order after selections have been made
- Ask the user if they have a loyalty card and apply 10% discount to final price if they do
- Display the final price (and if the user has a loyalty card, show them how much they've saved).

Pseudocode

```
DECLARE String firstName
DECLARE String surName
DECLARE String chosenTitle
DECLARE String displayName

SET firstName = method askFirstName
SET surName = method askSurName
SET chosenTitle = method askTitle
SET displayName = method getDisplayName(chosenTitle, firstName, surName)

DISPLAY displayName + ", welcome and thanks for choosing us!"

DECLARE int cutleryChoice

WHILE true DO
    TRY
        DISPLAY "Would you like some plastic cutlery with your order? Please choose:
                \n1 If you would \n2 If not
        SET cutleryChoice as userInput

        IF cutleryChoice = 1 THEN
            DISPLAY "Certainly, we'll bring it with your food."
            CLEAR leftover newline
            BREAK out of loop on valid input

        ELSE IF cutleryChoice = 2 THEN
            DISPLAY "No problem at all, we'll leave it out."
            CLEAR leftover newline
            BREAK out of loop on valid input

        ELSE
            DISPLAY "Sorry I didn't quite catch that, please choose option 1 or 2."
        END IF

    CATCH invalid input
        DISPLAY "Sorry, I don't know what you mean, please choose option 1 or 2."
        CLEAR invalid token
    END TRY
END WHILE

DISPLAY "Now let me show you our menu:"
DISPLAY "MENU"
DISPLAY "====="
```

```

DISPLAY "Starters:"
DISPLAY "====="
DISPLAY printStarterArray method
DISPLAY "====="
DISPLAY "Mains:"
DISPLAY "====="
DISPLAY printMainArray method
DISPLAY "====="
DISPLAY "Desserts:"
DISPLAY "====="
DISPLAY printDessertArray method
DISPLAY "====="
DISPLAY printDrinkArray method
DISPLAY "====="
DISPLAY "Special Offer: 10% discount with loyalty card!"
DISPLAY "You can also order a custom dish but these are more expensive and a
set price: £11.50 for starters, £15 for mains and £7.50 for desserts."
DISPLAY "=====

DECLARE int allergy

WHILE true DO
    TRY
        DISPLAY " Now before we start, are you allergic to any of the following: \n1 for
No \n2 for Garlic \n3 for Lentils \n4 for Seafood \n5 for Peanuts \n6 for Dairy \n7
for Other"

        SET allergy as userInput

        IF allergy < 1 OR allergy > 7 THEN
            DISPLAY "Please choose one of the options listed."
            CONTINUE to start of loop for retry
        END IF
        BREAK out of loop on valid input

    CATCH invalid input
        DISPLAY "Please enter a whole number from 1 to 7."
        CLEAR invalid token
    END TRY
END WHILE

SET boolean garlicAllergy = false
SET boolean lentilAllergy = false
SET boolean seafoodAllergy = false

```

SET boolean peanutAllergy = false

SET boolean dairyAllergy = false

SWITCH allergy

case 1: DISPLAY "Perfect, you can order anything from the menu!"

case 2: SET garlicAllergy = true

DISPLAY "garlicAllergy recorded"

case 3: SET lentilAllergy = true

DISPLAY "lentilAllergy recorded"

case 4: SET seafoodAllergy = true

DISPLAY "seafoodAllergy recorded"

case 5: SET peanutAllergy = true

DISPLAY "peanutAllergy recorded"

case 6: SET dairyAllergy = true

DISPLAY "dairyAllergy recorded"

case 7: DISPLAY "If you have any other allergies, I'm afraid these can't be
accommodated with our existing menu, but you can
order custom dishes to suit any dietary requirements."

END SWITCH

DISPLAY "So, let's begin your order!"

SET int orders = method askOrderCounter

CREATE Data Structure orderedItems to hold items users has ordered

CREATE Data Structure itemPrices to hold prices of items users has ordered

DECLARE int starter

DECLARE String starterName

DECLARE double starterPrice

FOR orderNumber FROM 0 TO orders - 1 DO

DISPLAY "Please place your order for order number " + (orderNumber + 1) + "!"

SET boolean starterConfirmed = false

WHILE true DO

WHILE true DO

TRY

```

        DISPLAY "For your starter, please choose from the following:"
        IF NOT garlicAllergy THEN
            DISPLAY "1 for Garlic Bread"

        END IF
        IF NOT lentilAllergy THEN
            DISPLAY "2 for Lentil Soup"
        END IF
        IF NOT seafoodAllergy THEN
            DISPLAY "3 for Prawn Cocktail"
        END IF
        DISPLAY "4 For Custom Dish (These are more expensive)"
        DISPLAY "5 To skip starter"
        SET starter = user input

        IF starter = 1 AND garlicAllergy = true THEN
            DISPLAY "Sorry, but you're allergic to this item so you can't order it."
            CONTINUE to start of loop to retry
        END IF
        IF starter = 2 AND lentilAllergy = true THEN
            DISPLAY "Sorry, but you're allergic to this item so you can't order it."
            CONTINUE to start of loop to retry
        END IF
        IF starter = 3 AND seafoodAllergy = true THEN
            DISPLAY "Sorry, but you're allergic to this item so you can't order it."
            CONTINUE to start of loop to retry
        END IF
        IF starter < 1 OR starter > 5 THEN
            DISPLAY "Please choose one of the allocated options."
            CONTINUE to start of loop to retry
        END IF
        BREAK out of inner loop if input is valid
    CATCH invalid input
        DISPLAY "Please choose one of the allocated options."
        CLEAR invalid token
    END TRY
END WHILE

CLEAR leftover newline

SWITCH starter
case 1: SET starterName = "Garlic Bread"
        SET starterPrice = 4.50

```

```

case 2: SET starterName = "Lentil Soup"
        SET starterPrice = 4.99
case 3: SET starterName = "Prawn Cocktail"
        SET starterPrice = 9.99
case 4: DISPLAY "Please enter the name of your custom starter"
        SET String customStarter = user input
        SET starterName = customStarter
        SET starterPrice = 11.50
case 5: SET starterName = "No starter"
        SET starterPrice = 0.00
END SWITCH

DISPLAY "You have chosen " + starterName + ": Please select: \n1 if you are happy with
        this and want to continue? \n2 if you would like to choose another option?"

DECLARE int starterConfirmation
WHILE true DO
    TRY
        starterConfirmation = user input

        IF starterConfirmation = 1 THEN
            DISPLAY "Great choice"
            SET starterConfirmed = true
            BREAK out of inner loop upon successful input
        ELSE IF starterConfirmation = 2 THEN
            DISPLAY "No problem, please choose your starter again"
            PRINT newline
            BREAK out of inner loop upon successful input
        ELSE
            DISPLAY "Please enter either 1 or 2"
        END IF
    CATCH invalid input
        DISPLAY "Please choose 1 or 2"
        CLEAR invalid token
    END TRY
END WHILE

    IF starterConfirmed = true
        ADD starterName to orderedItems
        ADD starterPrice to itemPrices
        BREAK out of outer loop
    END IF

```

```

    END outer WHILE loop
END FOR loop

DISPLAY "Great, your starter choices have been recorded. let's move on to the mains."

DECLARE int main
DECLARE String mainName
DECLARE double mainPrice

FOR orderNumber FROM 0 TO orders - 1 DO
    DISPLAY "Please place your order for order number " + (orderNumber + 1) + ":"
    SET boolean mainConfirmed = false

    WHILE true DO
        WHILE true DO
            TRY
                DISPLAY "For your main, please choose from the following:"
                IF NOT seafoodAllergy THEN
                    DISPLAY "1 For Fish and Chips"
                END IF
                IF NOT peanutAllergy THEN
                    DISPLAY "2 For Vegetarian Curry"
                END IF
                IF NOT dairyAllergy THEN
                    DISPLAY "3 For Cottage Pie"
                END IF
                DISPLAY "4 For Custom Dish (These are more expensive)"
                DISPLAY "5 To skip main course"
                SET main = user input

                IF main = 1 AND seafoodAllergy = true THEN
                    DISPLAY "Sorry, but you're allergic to this item so you can't order it."
                    CONTINUE to start of loop to retry
                END IF
                IF main = 2 AND peanutAllergy = true THEN
                    DISPLAY "Sorry, but you're allergic to this item so you can't order it."
                    CONTINUE to start of loop to retry
                END IF
                IF main = 3 AND dairyAllergy = true THEN
                    DISPLAY "Sorry, but you're allergic to this item so you can't order it."

```



```

        CONTINUE to start of loop to retry
    END IF

    IF main < 1 OR main > 5 THEN
        DISPLAY "Please choose one of the allocated options."
        CONTINUE to start of loop to retry
    END IF
    BREAK out of inner loop if input is valid
CATCH invalid input
    DISPLAY "Please enter a whole number from 1 to 5."
    CLEAR invalid token
END TRY
END WHILE

CLEAR leftover newline

SWITCH main
    case 1: SET mainName = "Fish and Chips"
            SET mainPrice = 12.50
    case 2: SET mainName = "Vegetarian Curry"
            SET mainPrice = 9.99
    case 3: SET mainName = "Cottage Pie"
            SET mainPrice = 11.50
    case 4: DISPLAY "Please enter the name of your custom main course"
            SET String customMain = user input
            SET mainName = customMain
            SET mainPrice = 15.00
    case 5: SET mainName = "No main"
            SET mainPrice = 0.00
END SWITCH

DISPLAY "You have chosen " + mainName + ": Please select: \n1 if you are happy with
        this and want to continue? \n2 if you would like to choose another option?"

DECLARE int mainConfirmation
WHILE true DO
    TRY
        mainConfirmation = user input

        IF mainConfirmation = 1 THEN
            DISPLAY "Great choice"
            SET mainConfirmed = true
            BREAK out of inner loop upon successful input
        
```

```

ELSE IF mainConfirmation = 2 THEN
    DISPLAY "No problem, please choose your main course again"
    PRINT newline
    BREAK out of inner loop upon successful input
ELSE
    DISPLAY "Please enter either 1 or 2"
END IF
CATCH invalid input
    DISPLAY "Please choose 1 or 2"
    CLEAR invalid token
END TRY
END WHILE

    IF mainConfirmed = true
        ADD mainName to orderedItems
        ADD mainPrice to itemPrices
        BREAK out of outer loop
    END IF
END outer WHILE loop
END FOR loop

DISPLAY "Excellent, your main course choices have been recorded. let's move on to the
dessert."

DECLARE int dessert
DECLARE String dessertName
DECLARE double dessertPrice

FOR orderNumber FROM 0 TO orders - 1 DO
    DISPLAY "Please place your order for order number " + (orderNumber + 1) + "
    SET boolean dessertConfirmed = false

    WHILE true DO

        WHILE true DO
            TRY
                DISPLAY "For your dessert, please choose from the following:"
                IF NOT dairyAllergy THEN
                    DISPLAY "1 For Trio of Ice-Cream"
                    DISPLAY "2 For Apple Pie"
                    DISPLAY " 3 For Cheesecake"
                END IF
            
```

```

    DISPLAY "4 For Custom Dish (These are more expensive)"
    DISPLAY "5 To skip DESSERT"
    SET dessert = user input

    IF dairyAllergy = true AND dessert ≠ equal 4 AND dessert ≠ 5 THEN
        DISPLAY "Sorry, but all our menu desserts contain dairy so you cannot order
                them with your allergy."
        CONTINUE to start of loop to retry
    END IF
    IF dessert < 1 OR dessert > 5 THEN
        DISPLAY "Please choose one of the allocated options."
        CONTINUE to start of loop to retry
    END IF
    BREAK out of inner loop if input is valid
    CATCH invalid input
        DISPLAY "Please enter a whole number from 1 to 5."
        CLEAR invalid token
    END TRY
END WHILE

CLEAR leftover newline

SWITCH dessert
    case 1: SET dessertName = "Trio of Ice-Cream"
            SET dessertPrice = 4.50
    case 2: SET dessertName = "Apple Pie"
            SET dessertPrice = 5.20
    case 3: SET dessertName = "Cheesecake"
            SET dessertPrice = 6.50
    case 4: DISPLAY "Please enter the name of your custom dessert"
            SET String customDessert = user input
            SET dessertName = customDessert
            SET dessertPrice = 7.50
    case 5: SET dessertName = "No dessert"
            SET dessertPrice = 0.00
END SWITCH

DISPLAY "You have chosen " + dessertName + ": Please select: \n1 if you are happy with
        this and want to continue? \n2 if you would like to choose another option?"

DECLARE int dessertConfirmation
WHILE true DO
    TRY

```

```

    dessertConfirmation = user input
IF dessertConfirmation = 1 THEN
    DISPLAY "Great choice"
    SET dessertConfirmed = true
    BREAK out of inner loop upon successful input
ELSE IF dessertConfirmation = 2 THEN
    DISPLAY "No problem, please choose your dessert again"
    PRINT newline
    BREAK out of inner loop upon successful input
ELSE
    DISPLAY "Please enter either 1 or 2"
END IF
CATCH invalid input
    DISPLAY "Please choose 1 or 2"
    CLEAR invalid token
END TRY
END WHILE

IF dessertConfirmed = true
    ADD dessertName to orderedItems
    ADD dessertPrice to itemPrices
    BREAK out of outer loop
END IF
END outer WHILE loop
END FOR loop

DISPLAY "Excellent, all your food choices have been recorded. Finally, what would you
    like to drink?"

DECLARE int drink
DECLARE String drinkName
DECLARE double drinkPrice

FOR orderNumber FROM 0 TO orders - 1 DO
    DISPLAY "Please place your order for order number " + (orderNumber + 1) + ":"
    SET boolean drinkConfirmed = false

    WHILE true DO

        WHILE true DO
            TRY
                DISPLAY "For your drink, please choose from the following:"

```

```

DISPLAY "1 For Tea"
DISPLAY "2 For Coffee"
DISPLAY "3 For Lemonade"
DISPLAY "4 For no drink"

SET drink = user input

IF drink < 1 OR drink > 4 THEN
    DISPLAY "Please choose one of the allocated options."
    CONTINUE to start of loop to retry
END IF
BREAK out of inner loop if input is valid
CATCH invalid input
    DISPLAY "Please enter a whole number from 1 to 4."
    CLEAR invalid token
END TRY
END WHILE

CLEAR leftover newline

SWITCH drink
case 1: SET drinkName = "Tea"
        SET drinkPrice = 2.99
case 2: SET drinkName = "Coffee"
        SET drinkPrice = 3.50
case 3: SET drinkName = "Lemonade"
        SET drinkPrice = 2.50
case 4: SET drinkName = "No drink"
        SET drinkPrice = 0.00
END SWITCH

DISPLAY "You have chosen " + drinkName + ": Please select: \n1 if you are happy
with this and want to continue?
\n2 if you would like to choose
another option?"

DECLARE int drinkConfirmation
WHILE true DO
    TRY
        drinkConfirmation = user input
        IF drinkConfirmation = 1 THEN
            DISPLAY "Great choice"
            SET drinkConfirmed = true
            BREAK out of inner loop upon successful input
        
```

```

    ELSE IF drinkConfirmation = 2 THEN
        DISPLAY "No problem, please choose your drink again"
        PRINT newline
        BREAK out of inner loop upon successful input
    ELSE
        DISPLAY "Please enter either 1 or 2"
    END IF
CATCH invalid input
    DISPLAY "Please choose 1 or 2"
    CLEAR invalid token
END TRY
END WHILE

IF drinkConfirmed = true
    ADD drinkName to orderedItems
    ADD drinkPrice to itemPrices
    BREAK out of outer loop
END IF
END outer WHILE loop
END FOR loop

DISPLAY "Your full order summary"

FOR each finalOrder IN orderedItems DO
    DISPLAY finalOrder
    PRINT newline
END FOR

DECLARE boolean loyaltyCard
SET loyaltyCard = method askLoyaltyCard

DECLARE finalPrice = 0

FOR each totalPrice IN itemPrices DO
    finalPrice = finalPrice + totalPrice
END FOR

IF loyaltyCard = true THEN
    SET double discount = finalPrice/10
    SET double finalPriceWithDiscount = finalPrice – discount
    DISPLAY "Dear displayName, your total to pay with loyalty card discount is:
        finalPriceWithDiscount And you have saved: discount"
ELSE
    DISPLAY "Dear displayName, your total to pay with no discount is: finalPrice"

```

```
END IF
END main
```

```
METHOD capitalise(input)
```

```
    IF input = null OR input = "" THEN
```

```
        RETURN input
```

```
    END IF
```

```
    DECLARE String firstLetter
```

```
    DECLARE String remainingLetters
```

```
    SET firstLetter = substring of input from index 0 to 1 converted to UPPERCASE
```

```
    SET remainingLetters = substring of input from index 1 to end converted to
```

```
        LOWERCASE
```

```
    RETURN firstLetter + remainingLetters
```

```
END METHOD
```

```
METHOD getDisplayName(title, firstName, surName)
```

```
    DECLARE String fullName
```

```
    SET fullName = capitalise(firstName) + " " + capitalise(surName)
```

```
    IF title = null OR title = "" THEN
```

```
        RETURN fullName
```

```
    END IF
```

```
    RETURN title + " " + fullName
```

```
END METHOD
```

```
METHOD askFirstName (scanner)
```

```
    DECLARE String firstName
```

```
    WHILE true DO
```

```
        DISPLAY "Hello and welcome to our online restaurant ordering service, what is your  
            first name?"
```

```
        SET firstName = user input
```

```
        IF firstName is EMPTY THEN
```

```
            DISPLAY "Name can't be blank - Please enter your name."
```

```
        CONTINUE to start of loop
    END IF

    IF firstName contains characters other than letters or spaces THEN
        DISPLAY "Please use letters and spaces only."
        CONTINUE to start of loop
    END IF
    RETURN firstName
END WHILE
END METHOD
```

```
METHOD askSurName (scanner)
```

```
    DECLARE String surName

    WHILE true DO
        DISPLAY "And what is your surname?"
        SET surName = user input

        IF surName is EMPTY THEN
            DISPLAY "Surname can't be blank - Please enter your name."
            CONTINUE to start of loop
        END IF

        IF SurName contains characters other than letters or spaces THEN
            DISPLAY "Please use letters and spaces only."
            CONTINUE to start of loop
        END IF
        RETURN surName
    END WHILE
END METHOD
```

```
METHOD askTitle (scanner)
```

```
    DECLARE String title

    WHILE true DO
        DISPLAY "And what is your title? Please choose from Mr, Mrs, Miss, or Other."
        SET title = user input

        IF title is EMPTY THEN
            DISPLAY "Please choose Mr, Mrs, Miss, or Other."
            CONTINUE to start of loop
        END IF
```



```

IF title contains characters other than letters or spaces THEN
    DISPLAY "Please use letters and spaces only."
    CONTINUE to start of loop
END IF

IF title ≠ "Mr" AND title ≠ "Mrs" AND title ≠ "Miss" AND title ≠ "Other" THEN
    DISPLAY "Please choose Mr, Mrs, Miss, or Other."
    CONTINUE to start of loop
END IF

BREAK out of loop
END WHILE

SWITCH title
CASE "Mr":
    RETURN "Mr"
CASE "Mrs"
    RETURN "Mrs"
CASE "Miss"
    RETURN "Miss"
CASE "Other"
    DISPLAY "Please enter your preferred title, if you don't want one, leave this blank."
    DECLARE String customTitle
    SET customTitle = user input
    RETURN customTitle
DEFAULT
    DISPLAY "Please choose one of the aforementioned options."
"
    RETURN ""
END SWITCH
END METHOD

```

METHOD askOrderCounter (scanner)

```

DECLARE int orders

```

```

WHILE true DO

```

```

    TRY

```

```

        DISPLAY "How many orders are you placing today"

```

```

        orders = user input

```

```

IF orders < 1 OR orders > 5 THEN
    DISPLAY "Sorry we can't fulfill your request, there needs to be a minimum of 1
            order and maximum of 5."
    CONTINUE to start of loop
END IF
DISPLAY "Perfect you'd like to make" + orders "orders"
RETURN orders
CATCH invalid input
    DISPLAY "Please enter a whole number between from 1 to 5."
    CLEAR bad token
END TRY
END WHILE
END METHOD

```

```

METHOD askLoyaltyCard (scanner)

```

```

    WHILE true DO
        TRY
            DISPLAY "Do you have a loyalty card with us? \n1 = Yes \n2 = No"
            DECLARE int loyaltyChoice
            SET loyaltyChoice = user input

            IF loyaltyChoice = 1 THEN
                DISPLAY "Excellent, you'll have a 10% discount applied to your final price, thanks
                        for sticking with us!"
                RETURN true
            ELSE IF loyaltyChoice = 2 THEN
                DISPLAY "No worries, you'll be charged full price for your order."
            END IF
        END TRY
    END WHILE
END METHOD

```

```
        RETURN false
    ELSE
        DISPLAY "Please choose either 1 or 2."
    END IF
CATCH invalid input
    DISPLAY "Please choose 1 or 2."
    CLEAR bad token
END TRY
END WHILE
END METHOD
```

METHOD printStarterArray

```
    DECLARE data structure starterMenu
    FOR EACH starterItems IN starterMenu DO
        DISPLAY starterItems
    END FOR
END METHOD
```

METHOD printMainArray

```
    DECLARE data structure mainMenu
    FOR EACH mainItems IN mainMenu DO
        DISPLAY mainItems
    END FOR
END METHOD
```

METHOD printDessertArray

```
    DECLARE data structure dessertMenu
    FOR EACH dessertItems IN dessertMenu DO
```

```
    DISPLAY dessertItems  
END FOR  
END METHOD
```

```
METHOD printDrinkArray  
  
    DECLARE data structure drinkMenu  
    FOR EACH drinkItems IN drinkMenu DO  
        DISPLAY drinkItems  
    END FOR  
END METHOD
```